1)

$E_{batt} = 50\,kWh$ $\quad L_{cable} = 2 \times 3\,m$

$V_{batt} = 100\,V$ $\quad D_{cable} = 1\,cm$

$t_{charge} = 0{,}25\,h$ $\quad A_{cable} = \frac{1}{4}\pi d^2 \approx 0{,}785\,cm^2$

a) $\frac{50\,kWh}{100\,V} = 500\,Ah$

$\frac{500\,Ah}{0{,}25\,h} = 2000\,A = 2\,kA$

b) $R = \rho \frac{L}{A} = 1{,}68 \cdot 10^{-8}\,\Omega m \cdot \frac{6\,m}{0{,}785\,cm^2} = 51{,}97\,\mu\Omega$

$U_C = RI = 51{,}97\,\mu\Omega \cdot 2\,kA = 0{,}1\,V$

$P_C = UI = 0{,}1\,V \cdot 2\,kA = 207{,}6\,W$

c) $E_C = P_C \cdot T_{charge} = 207{,}5\,W \cdot 0{,}25\,h = 51{,}97\,Wh$

$\frac{E_C}{E_C + E_{batt}} = \frac{51{,}97}{51{,}97 + 50\,kWh} = 0{,}001\,\%$

d) $\frac{50\,kWh}{300\,V} = 100\,Ah$

$\frac{100\,Ah}{0{,}25\,h} = 400\,A$

$P_{C400} = RI^2 = 51{,}97\,\mu\Omega\,(400\,A)^2 = 8{,}31\,W$

$E_{C400} = P_{C400} \cdot T_C = 8{,}31\,W \cdot 0{,}25\,h = 2{,}08\,Wh$

$\frac{E_{C400}}{E_{C400} + E_{batt}} = 0{,}00004\,\%$

2)

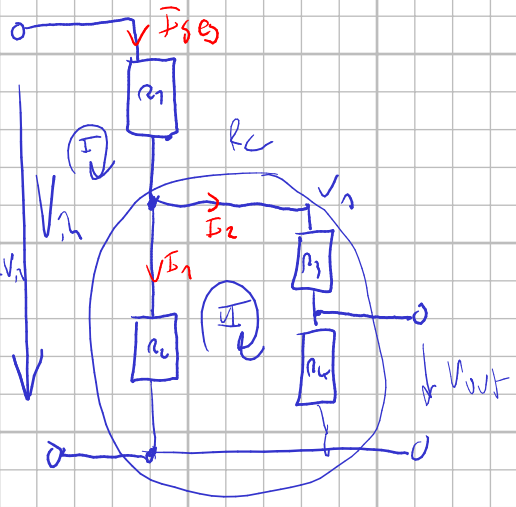
$R_1 = 1$

a) I $V_{out} = V_1 \cdot \frac{R_{ges}}{R_2 + R_3} = \frac{1}{2} V_1$

$R_{ges} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_3 + R_4}} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$

II $V_1 = V_{in} \frac{R_{ges}}{R_1 + R_{ges}} = \frac{\frac{2}{3}}{1 + \frac{2}{3}} = \frac{\frac{2}{3}}{\frac{5}{3}} = \frac{2}{5} V_{in}$

in I $V_{out} = \frac{1}{2} \cdot \frac{2}{5} V_{in} = \frac{1}{5} V_{in}$



b) $I_1 = \frac{V_1}{R_2}$

$I_2 = \frac{V_1}{R_3 + R_4}$

I $V_{in} = I_{ges} R_1 + V_1 = (I_1 + I_2) R_1 + V_1$

$= \left(\frac{V_1}{R_2} + \frac{V_1}{R_3 + R_4}\right) R_1 + V_1 = V_1 + \frac{V_1}{2} + V_1 = \frac{5}{2} V_1$

$\Rightarrow V_1 = \frac{2}{5} V_{in}$

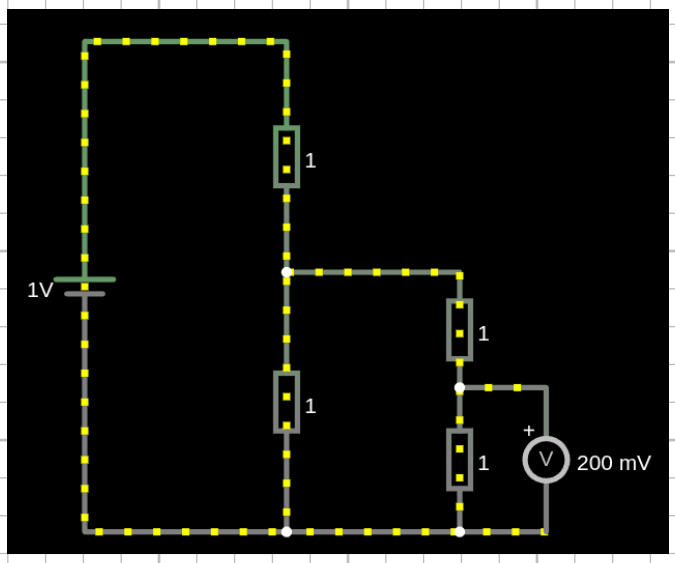
II $V_{out} = V_1 - R_3 I_2$

$= V_1 - R_3 \frac{V_1}{R_3 + R_4}$

$= V_1 - \frac{1}{2} V_1$

$= \frac{1}{2} V_1$

$V_{out} = \frac{1}{2} \cdot \frac{2}{5} V_{in} = \frac{1}{5} V_{in}$



3)

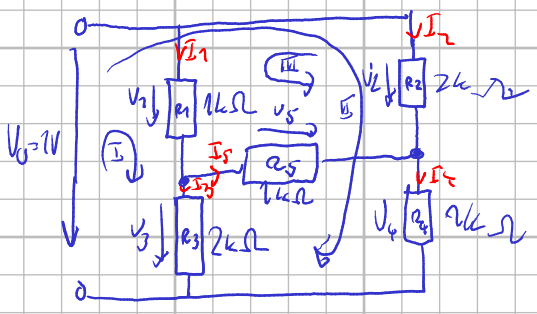
a) I: $V_1 + V_3 = V_0 \Rightarrow R_1 I_1 + R_3 I_3 = 1$

II: $V_2 + V_4 = V_0 \Rightarrow R_2 I_2 + R_4 I_4 = 1$

III: $V_5 + V_1 - V_2 = 0 \Rightarrow R_5 I_5 + R_1 I_1 - R_2 I_2 = 0$

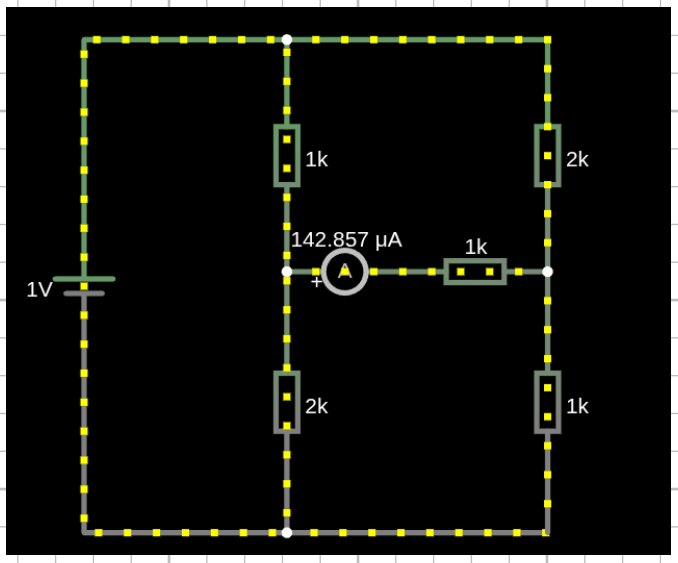
IV: $I_1 - I_3 - I_5 = 0$

V: $I_2 + I_5 - I_4 = 0$

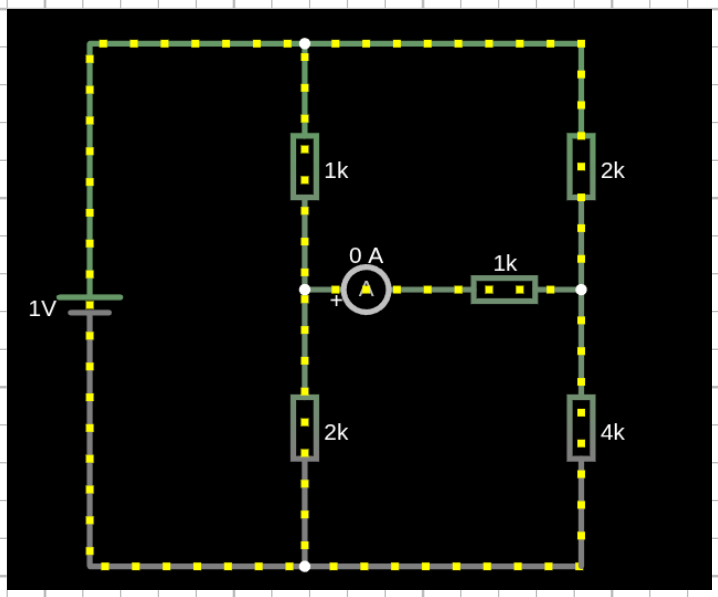


$$\begin{pmatrix} R_1 & 0 & R_3 & 0 & 0 \\ 0 & R_2 & 0 & R_4 & 0 \\ R_1 & -R_2 & 0 & 0 & R_5 \\ 1 & 0 & -1 & 0 & -1 \\ 0 & 1 & 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Solve: $I_1 = 428{,}57\ \mu A$
$I_2 = 285{,}71\ \mu A$
$I_3 = 285{,}71\ \mu A$
$I_4 = 428{,}57\ \mu A$
$I_5 = 142{,}86\ \mu A$



b) $V_1 = V_1$ wenn $\frac{R_1}{R_3} = \frac{R_2}{R_4} \Rightarrow R_4 = R_2 \frac{R_3}{R_1} = 2k\Omega \cdot 2 = 4k\Omega$



4)

a) $\frac{dV}{dt} = \frac{I}{C} = \frac{1\mu A}{3nF} = 333{,}34\ \frac{V}{s}$

$\frac{3{,}52\,V}{10{,}56\,ms} = 333{,}34\ \frac{V}{s}$ Steigung in der Simulation

b) $I = \frac{dV}{dt} C = 0{,}2\ \frac{kV}{s} \cdot 3nF = 0{,}6\,\mu A$

$R = \frac{V}{I} = \frac{3V}{0{,}6\mu A} = 5M\Omega$